

Lecture - 01/15/2025

↳ Midterm: Mar. 10th in class

→ start: 11:30 - 2:00

↳ lifecycle model:

→ $\beta = 1; r = 0$

→ $u'(c_t) = \beta(1+r)u'(c_{t+1}) \Rightarrow c_t = c_{t+1} = c$

↳ generic lifetime b.c.

$$\rightarrow \sum_{t=0}^T \frac{c_t}{(1+r)^t} = a_0 + \sum_{t=1}^N \frac{y}{(1+r)^t} + \sum_{N+1}^T \frac{0}{(1+r)^{t+1}}$$

→ $Tc = Ny$

$$\rightarrow c = T^{-1} \cdot \underbrace{Ny}_{\substack{\text{lifetime} \\ \text{human } W}}$$

→ equally spread wealth over lifetime

↳ $t=1: y, c \Rightarrow \text{save } y-c \Rightarrow \text{fin } W: y-c$
 $=2: y, c \Rightarrow \text{save } y-c \Rightarrow \text{fin } W: 2(y-c)$
 $\vdots$
 $=N: y, c \Rightarrow \text{save } y-c \Rightarrow \text{fin } W: N(y-c)$

↳ define a_t as holdings at time t . s.t. $t \leq N$

$$\rightarrow a_t = t(y - c) = t\left(y - \frac{N}{T}y\right) = ty\left(\frac{T-N}{T}\right)$$

↳ remaining human wealth:

$$\rightarrow (N-t)y$$

↳ consuming pieces of pie as go

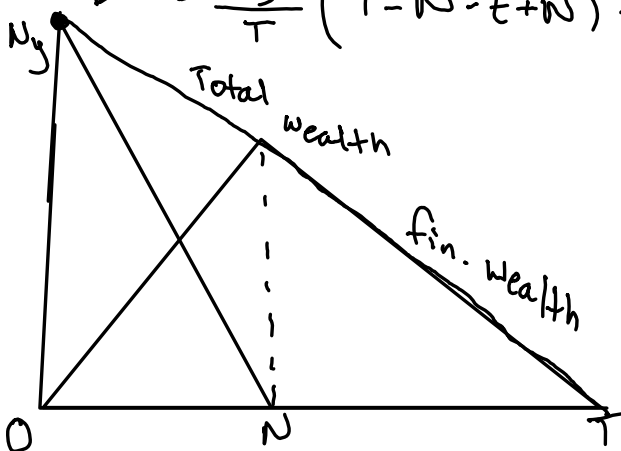
$\rightarrow$ as $t \rightarrow N$, human wealth $\rightarrow 0$.

$$\rightarrow a_N = Ny \frac{T-N}{T}$$

↳ decumulate by c each period

$$\rightarrow a_t = \underbrace{Ny \frac{T-N}{T}}_{a_N} - \underbrace{\frac{N}{T}y}_{c}(t-N); t > N$$

$$\rightarrow = \frac{Ny}{T} (T-N-t+N) = \frac{Ny}{T} (T-t)$$



$$\begin{aligned}
 & \xrightarrow{\text{rem human w.}} \xrightarrow{\text{fin w.}} (N-t)y + \frac{T-N}{T} ty \\
 & = Ny + yt \left[\frac{T-N}{T} - 1 \right] = Ny - \frac{N}{T} yt = yN \left(1 - \frac{t}{T} \right)
 \end{aligned}$$

Consumption and saving
w/ uncertainty

$$\xrightarrow{\max} \{c_t\}_{t=0}^{\infty} \quad E_0 \sum_{t=0}^{\infty} \beta^t u(c_t)$$

$$\text{s.t. } c_t + a_{t+1} = y_t + a_t(1+r)$$

$$\frac{a_t}{(1+r)^T} \xrightarrow{T \rightarrow \infty} 0$$

y iid

→ Euler:

$$u'(c_t) = \beta(1+r) E_t u'(c_{t+1})$$

$$\xrightarrow{\text{Assume } \beta(1+r)=1}$$

$$\rightarrow u'(c_t) = E_t u'(c_{t+1})$$

$$\begin{aligned}
 \star & \xrightarrow{\text{Assume further that } u(c) = \alpha c - \gamma c^2 \cdot 2^{-1} \star} \\
 & \text{s.t. } \alpha, \gamma > 0 \\
 & \rightarrow \text{quadratic } u(c)
 \end{aligned}$$

$$\hookrightarrow u'(c) = \alpha - \gamma c$$

$$\hookrightarrow \alpha - \gamma c_t = E_t(\cancel{\alpha - \gamma c_{t+1}})$$

$$\hookrightarrow -\gamma c_t = -\gamma E_t c_{t+1} \Rightarrow c_t = E_t c_{t+1}$$

$$\hookrightarrow \exists x \Rightarrow x_{t+1} = \overbrace{E_t x_{t+1}}^{\text{con. exp.}} + \underbrace{\varepsilon_{t+1}}_{\text{shock}}$$

$$\hookrightarrow c_{t+1} = E_t c_{t+1} + \varepsilon_{t+1}$$

$$\rightarrow c_{t+1} = c_t + \varepsilon_{t+1} \Rightarrow c \text{ follows random walk}$$

$$\rightarrow c_{t+1} - c_t = \varepsilon_{t+1}$$

$$\rightarrow \Delta c_{t+1} = \varepsilon_{t+1}$$

} all equivalent

$$\hookrightarrow E \varepsilon_{t+1} = 0$$

$\rightarrow$ consumption changes are unpredictable

Derive Lifetime B.C.

$$\hookrightarrow a_{t+1} + c_t = a_t(1+r) + y_t$$

$$\rightarrow t=0: a_1 + c_0 = a_0 + y_0$$

$$a_0 = a_1 + c_0 - y_0$$

$$a_0 = \frac{a_1}{(1+r)} + \frac{c_1 - y_1}{(1+r)} + c_0 - y_0 \Rightarrow a_0 = \frac{a_2}{(1+r)^2} + \frac{c_2 - y_2}{(1+r)^2} + c_0 - y_0$$

$$\rightarrow t=1: a_2 + c_1 = a_1(1+r) + y_1$$

$$a_1 = \frac{a_2}{(1+r)} + \frac{c_1 - y_1}{(1+r)}$$

$$\rightarrow t=2: a_3 + c_2 = a_2(1+r) + y_2$$

$$a_2 = \frac{a_3}{(1+r)} + \frac{c_2 - y_2}{(1+r)}$$

$$\rightarrow t=T: a_0 = \frac{a_{T+1}}{(1+r)^T} + \sum_{t=0}^T \frac{c_t - y_t}{(1+r)^t}$$

$$\rightarrow t=\infty: \frac{a_{T+1}}{(1+r)^T} \rightarrow 0 \therefore \sum_{t=0}^{\infty} \frac{c_t}{(1+r)^t} = a_0 + \sum_{t=0}^{\infty} \frac{y_t}{(1+r)^t}$$

$$\hookrightarrow E_0 \sum_{t=0}^{\infty} \frac{c_t}{(1+r)^t} = a_0 + E_0 \sum_{t=0}^{\infty} \frac{y_t}{(1+r)^t}$$

Solve for C:

$$\hookrightarrow E_0 c_t = E_0 \underbrace{E_{t-1} c_t}_{\text{law of iterated expectation}}$$

$$\rightarrow E_0 c_t = E_0 c'_{t-1} = E_0 E_{t-2} c_{t-1} = E_0 c_{t-2} = E_0 c_1 = c_0$$